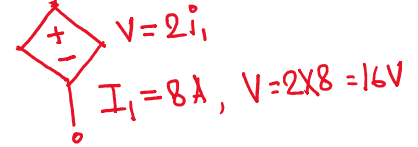
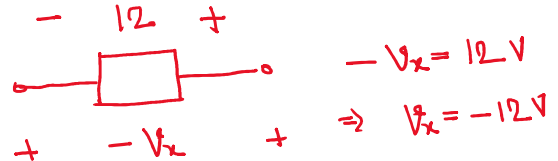
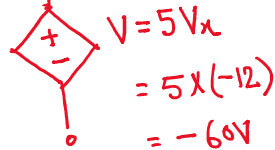


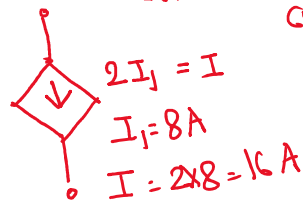
Controlled by current



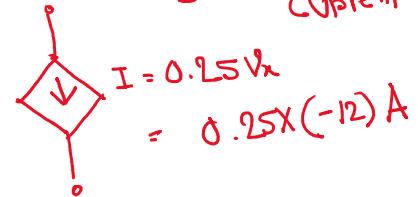
Controlled by voltage



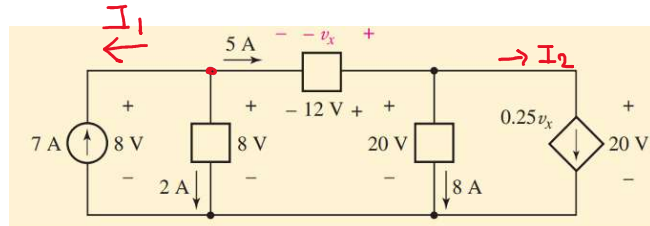
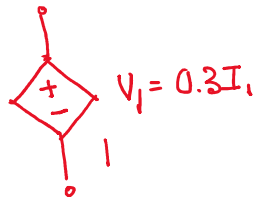
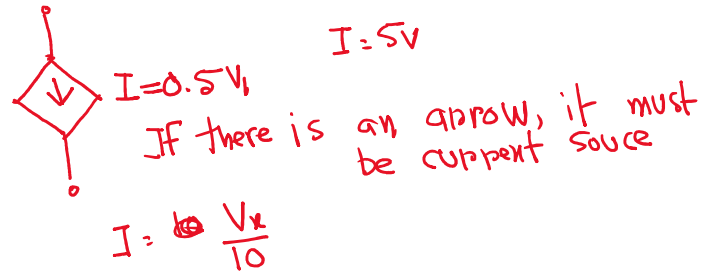
Current controlled current source



Voltage controlled current source



+ must be voltage source



$I_1 = -7A$, I have assigned current so that it appears to be entering the +ve terminal.

$$P_1 = V_1 I_1 = 8V \times (-7A) = -56W \text{ (supplying)}$$

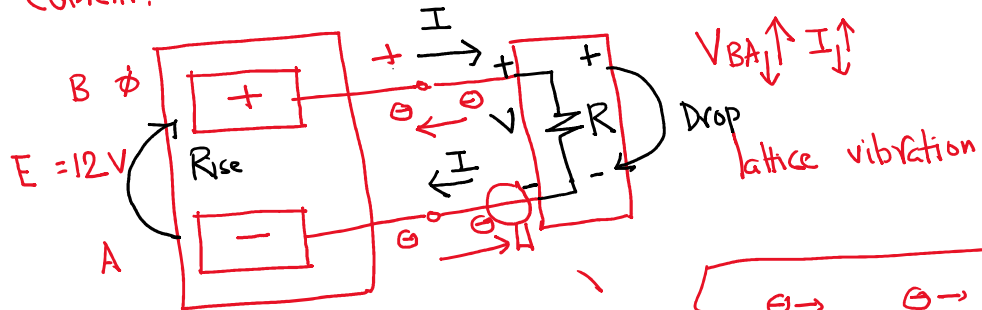
$$I_2 = 0.25V_x = 0.25 \times (-12)A$$

$$P_2 = V_2 I_2 = 20V \times 0.25 \times (-12)W$$

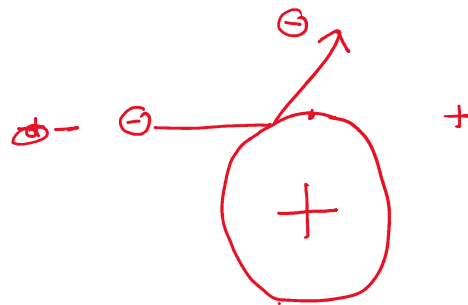
- Voltage/potential diff.
- Current.

$$P_2 = V_2 I_2 = 20 \times 0.25 \times (-12) \text{ W}$$

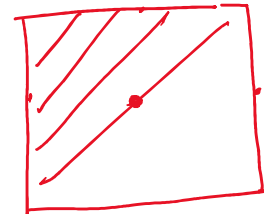
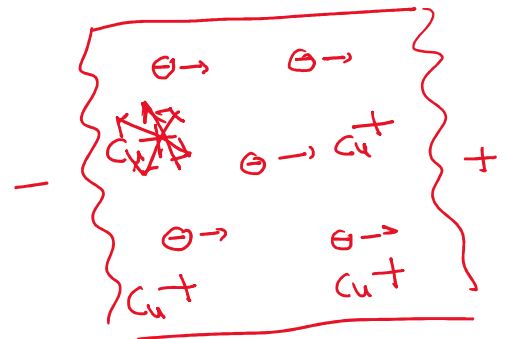
(supplying)



electrons experiences opposition.



movement of electron is limited by the opposition



Voltage
(Cause)

Current
(Response)

$$\text{Response} = \frac{\text{Cause}}{\text{Opposition}}$$

$$\text{Current} = \frac{\text{Voltage}}{\text{Resistance}}$$

$$I = \frac{E}{R}, \quad R: \text{ohm } (\Omega)$$

$$\Rightarrow E = IR \text{ [ohm's law]}$$

Conductor: $R=0$

Power supplied = Power Absorbed

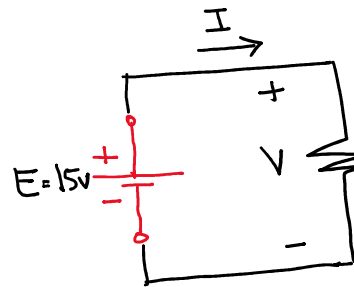
$$EI = VI \Rightarrow E = V$$

Voltage Rise
(- → +)

Voltage drop
(+ → -)

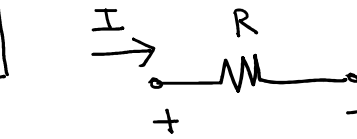
Voltage Rise
(- → +)

Voltage drop
(+ → -)



$$V = E = 15V$$

$$I = \frac{E}{R} = \frac{V}{R} = \frac{15}{10} = 1.5A$$



$$I = \frac{V_{AB}}{R} = \frac{3V}{20} A$$

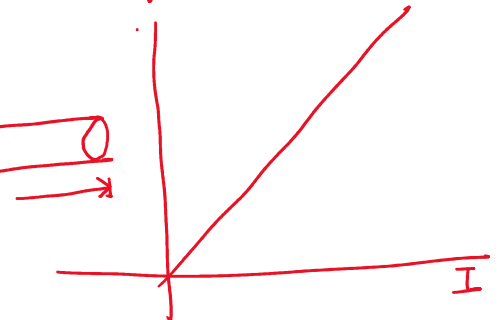
$$V = IR, V \propto I$$

$$y = mx$$

$$R \propto L$$

$$\propto \frac{1}{A}$$

$$R = \rho \frac{L}{A}$$



ρ : Resistivity

$$\text{Conductivity} = \frac{1}{\text{Resistivity}}$$

$$\frac{V}{I} = R$$

$$\Rightarrow \frac{I}{V} = \frac{1}{R} = G$$

$$\sigma = \frac{1}{\rho}$$

$$1S = \frac{1A}{1V} = 1\Omega^{-1}$$

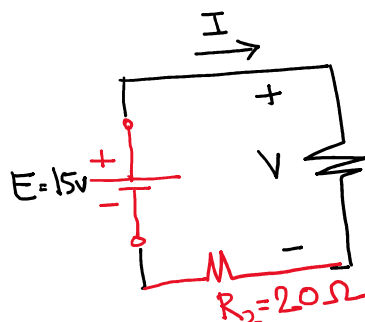
Siemens

R: Ω (ohm)

G: Ω^{-1} (mho)

$$P = VI = IRI = I^2R$$

$$= V \cdot \frac{V}{R} = \frac{V^2}{R}$$



$$I = \frac{E}{R}$$

- ohm's law alone isn't enough

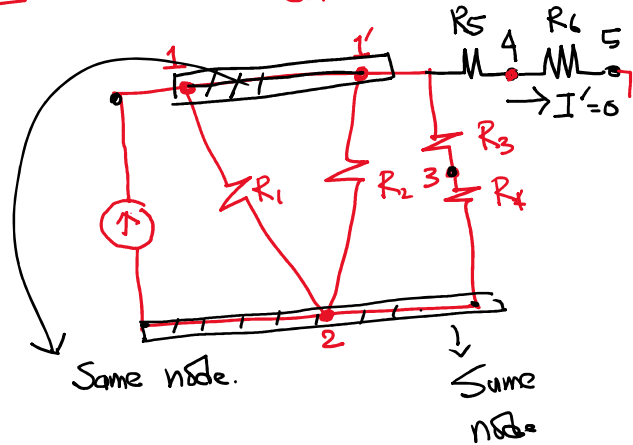
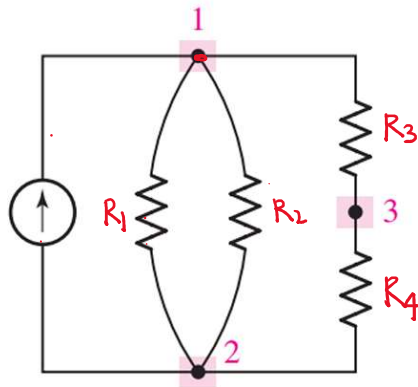
- KVL: Kirchhoff Voltage law
KCL: " Current law

or more

$$R_2 = 20 \Omega$$

KCL: $\sum u$

Node: or more
More than 2 elements are
connected is node

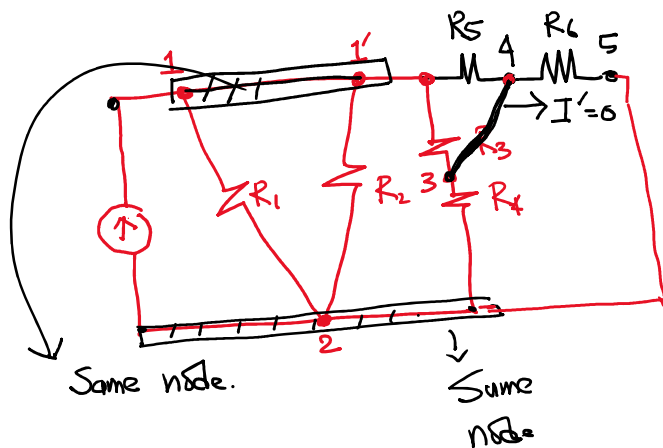


Voltage are same

→ the point must belong to a closed network (ckt) to be considered as valid node.
(Floating node)

→ Current flowing through floating node is zero.

$$V_{RS} = R_5 \times I' = 0V$$



If you start from
Point 4 you can come
back to it

V_3 : Node 3 voltage
 V_4 : Node 4 voltage
 $V_3 = V_4$